ECAS Summer Course

Inference for Quantile Regression

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La Roche-en-Ardennes: September, 2005

Inference for Quantile Regression

- Sample Quantiles and Prediction Intervals
- Wald tests for Quantile Regression
- Classical Rank Tests and the Quantile Regression Dual
- Inference on the Quantile Regression Process I
- Inference on the Quantile Regression Process II

Asymptotics for the Sample Quantiles

Minimizing $\sum_{i=1}^{n} \rho_\tau(y_i - \xi)$ consider

$$g_n(\xi) = -\sum_{i=1}^{n} \psi_\tau(y_i - \xi) = \sum_{i=1}^{n} (I(y_i < \xi) - \tau).$$

By convexity of the objective function,

$$\{\hat{\xi}_\tau > \xi\} \Leftrightarrow \{g_n(\xi) < 0\}.$$

This leads to distribution-free prediction intervals for the sample quantiles based on adjacent order statistics. The DeMoivre-Laplace CLT yields, expanding F,

$$\sqrt{n}(\hat{\xi}_\tau - \xi) \rightsquigarrow \mathcal{N}(0, \omega^2(\tau, F))$$

where $\omega^2(\tau, F) = \tau(1-\tau)/f^2(F^{-1}(\tau))$. The Bahadur-Kiefer representation theory provides further refinement of this result.

Asymptotic Theory of Quantile Regression I

In the classical linear model,

$$y_i = x_i\beta + u_i$$

with u_i iid from dfF, with density $f(u) > 0$ on its support $\{u|0 < F(u) < 1\}$, the joint distribution of $\sqrt{n}(\hat{\beta}_n(\tau_i) - \beta(\tau_i))_{i=1}^m$ is asymptotically normal with mean 0 and covariance matrix $\Omega \otimes D^{-1}$. Here $\beta(\tau) = \beta + F_u^{-1}(\tau)e_1, e_1 = (1, 0, \ldots, 0)'$, $x_{1i} \equiv 1, n^{-1}\sum x_i x_i' \to D$, a positive definite matrix, and

$$\Omega = ((\tau_i \wedge \tau_j - \tau_i\tau_j)/(f(F^{-1}(\tau_i))f(F^{-1}(\tau_j)))_{i,j=1}^m.$$

Asymptotic Theory of Quantile Regression II

When the response is conditionally independent over i, but not identically distributed, the asymptotic covariance matrix of $\zeta(\tau) = \sqrt{n}(\hat{\beta}(\tau) - \beta(\tau))$ is somewhat more complicated. Let $\xi_i(\tau) = x_i\beta(\tau)$, $f_i(\cdot)$ denote the corresponding conditional density, and define,

$$\begin{aligned} J_n(\tau_1, \tau_2) &= (\tau_1 \wedge \tau_2 - \tau_1\tau_2) n^{-1} \sum_{i=1}^{n} x_i x_i', \\ H_n(\tau) &= n^{-1} \sum x_i x_i' f_i(\xi_i(\tau)). \end{aligned}$$

Under mild regularity conditions on the $\{f_i\}$'s and $\{x_i\}$'s, we have joint asymptotic normality for $(\zeta(\tau_i), \ldots, \zeta(\tau_m))$ with the (sandwich) covariance matrix

$$V_n = (H_n(\tau_i)^{-1} J_n(\tau_i, \tau_j) H_n(\tau_j)^{-1})_{i,j=1}^{m}.$$

Wald Tests for Quantile Regression

Tests of linear and nonlinear hypothesis on one or more vectors of quantile regression coefficients can be easily made based on an estimated version of the sandwich form of the covariance matrix.

The only complication of this strategy is the need to estimate the nuisance parameters, $f_i(F^{-1}_{Y|x_i}(\tau|x_i))$ for $i = 1, 2, \ldots, n$. This is most simply done using the Siddiqui estimator,

$$f_i(F^{-1}_{Y|x_i}(\tau|x_i)) = \frac{2h_n}{x_i^\top(\hat{\beta}(\tau+h) - \hat{\beta}(\tau-h))}$$

Choice of bandwidth is still a subject of ongoing research. The default choice in my R package `quantreg` is based on work of Sheather and Hall.

Rank Based Inference for Quantile Regression

- Ranks play a fundamental *dual* role in QR inference.
- Classical rank tests for the p-sample problem extended to regression
- Rank tests play the role of Rao (score) tests for QR.

Two Sample Location-Shift Model

$$X_1, \ldots, X_n \sim F(x) \qquad \text{“Controls”}$$
$$Y_1, \ldots, Y_m \sim F(x-\theta) \qquad \text{“Treatments”}$$

Hypothesis:

$$H_0: \qquad \theta = 0$$
$$H_1: \qquad \theta > 0$$

The Gaussian Model $F = \Phi$

$$T = (\bar{Y}_m - \bar{X}_n)/\sqrt{n^{-1} + m^{-1}}$$

UMP Tests:

$$\text{critical region } \{T > \Phi^{-1}(1-\alpha)\}$$

Wilcoxon-Mann-Whitney Rank Test

Mann-Whitney Form:

$$S = \sum_{i=1}^{n} \sum_{j=1}^{m} I(Y_j > X_i)$$

Heuristic: If treatment responses are larger than controls for most pairs (i, j), then H_0 should be rejected.

Wilcoxon Form: Set $(R_1, \ldots, R_{n+m}) = \text{Rank}(Y_1, \ldots, Y_m, X_1, \ldots X_n)$,

$$W = \sum_{j=1}^{m} R_j$$

Proposition: $S = W - m(m+1)/2$ so Wilcoxon and Mann-Whitney tests are equivalent.

Pros and Cons of the Transformation to Ranks

Thought One:

Gain: Null Distribution is independent of F.

Loss: Cardinal information about data.

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Gain: Null Distribution is independent of F.

Loss: Cardinal information about data.

Thought Two:

Gain: Student t-test has quite accurate size provided $\sigma^2(F) < \infty$.

Loss: Student t-test uses cardinal information badly for long-tailed F.

Asymptotic Relative Efficiency of Wilcoxon versus Student t-test

Pitman Alternatives: $H_n : \theta_n = \theta_0/\sqrt{n}$

(t-test)$^2 \rightsquigarrow \chi_1^2(\theta_0^2/\sigma^2(F))$

(Wilcoxon)$^2 \rightsquigarrow \chi_1^2(12\theta_0^2(\int f^2)^2)$

ARE(W, t, F) $= 12\sigma^2(\mathrm{F})[\int f^2(x)dx]^2$

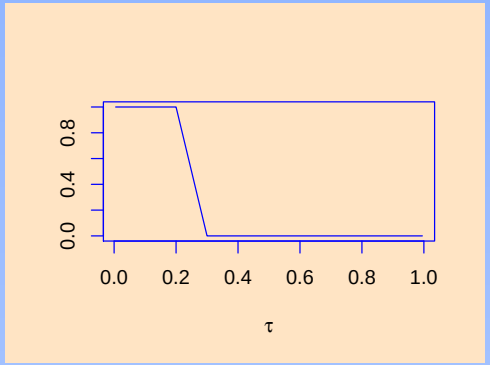
F	N	U	Logistic	DExp	LogN	t_2
ARE	.955	1.0	1.1	1.5	7.35	∞

Theorem (Hodges-Lehmann) For all F, $ARE(W, t, F) \geq .864$.

Hájek 's Rankscore Generating Functions

Let $Y_1, \ldots, Y_n$ be a random sample from an absolutely continuous df F with associated ranks $R_1, \ldots, R_n$, Hájek 's rank generating functions are:

$$\hat{a}_i = \begin{cases} 1 & \text{if } t \leq (R_i - 1)/n \\ R_i - tn & \text{if } (R_i - 1)/n \leq t \leq R_i/n \\ 0 & \text{if } R_i/n \leq t \end{cases}$$



Linear Rank Statistics Asymptotics

Theorem (Hájek (1965)) Let $c_n = (c_{1n}, \ldots, c_{nn})$ be a triangular array of real numbers such that

$$\max_i (c_{in} - \bar{c}_n)^2 / \sum_{i=1}^{n} (c_{in} - \bar{c}_n)^2 \to 0.$$

Then

$$Z_n(t) = (\sum_{i=1}^{n} (c_{in} - \bar{c}_n)^2)^{-1/2} \sum_{j=1}^{n} (c_{jn} - \bar{c}_n) \hat{a}_j(t)$$

converges weakly to a Brownian Bridge.

Some Asymptotic Heuristics

The Hájek functions are approximately indicator functions

$$\hat{a}_i(t) \approx I(Y_i > F^{-1}(t)) = I(F(Y_i) > t)$$

Since $F(Y_i) \sim U[0,1]$, linear rank statistics may be represented as

$$\int_0^1 \hat{a}_i(t) d\varphi(t) \approx \int_0^1 I(F(Y_i) > t) d\varphi(t) = \varphi(F(Y_i)) - \varphi(0)$$

$$\begin{aligned} \int_0^1 Z_n(t) d\varphi(t) &= \sum w_i \int \hat{a}_i(t) d\varphi(t) \\ &= \sum w_i \varphi(F(Y_i)) + o_p(1). \end{aligned}$$

Duality of Ranks and Quantiles

Quantiles may be *defined* as

$$\hat{\xi}(\tau) = argmin \sum \rho_\tau(y_i - \xi)$$

where $\rho_\tau(u) = u(\tau - I(u < 0))$. This can be formulated as a linear program whose dual solution

$$\hat{a}(\tau) = argmax\{y'a | 1_n' a = (1-\tau)n, a \in [0,1]^n\}$$

generates the Hájek rankscore functions.

Reference: Gutenbrunner and Jurečková (1992).

Regression Quantiles and Rank Scores:

$$\hat{\beta}_n(\tau) = \arg\min_{b \in R^p} \sum \rho_\tau(y_i - x_i' b)$$

$$\hat{a}_n(\tau) = \arg\max_{a \in [0,1]^n} \{y' a | X' a = (1-\tau) X' 1_n\}$$

$x'\hat{\beta}_n(\tau)$ Estimates $Q_Y(\tau|x)$
Piecewise constant on $[0,1]$.
For $X = 1_n$, $\hat{\beta}_n(\tau) = \hat{F}_n^{-1}(\tau)$.

$\{\hat{a}_i(\tau)\}_{i=1}^n$ Regression rankscore functions
Piecewise linear on $[0,1]$.
For $X = 1_n$, $\hat{a}_i(\tau)$ are the Hajek rank generating functions.

Regression Rank Tests

$$Y = X\,\beta + Z\,\gamma + u$$

$$H_0 : \gamma = 0 \text{ versus } H_n : \gamma = \gamma_0/\sqrt{n}$$

Given the regression rank score process for the restricted model,

$$\hat{a}_n(\tau) = \arg\max\{Y'a \,|\, X'a = (1-\tau)X'1_n\}$$

A test of H_0 is based on the linear rank statistics,

$$\hat{b}_n = \int_0^1 \hat{a}_n(t)\, d\varphi(t)$$

Choice of the score function φ permits test of location, scale or (potentially) other effects.

Regression Rank Tests

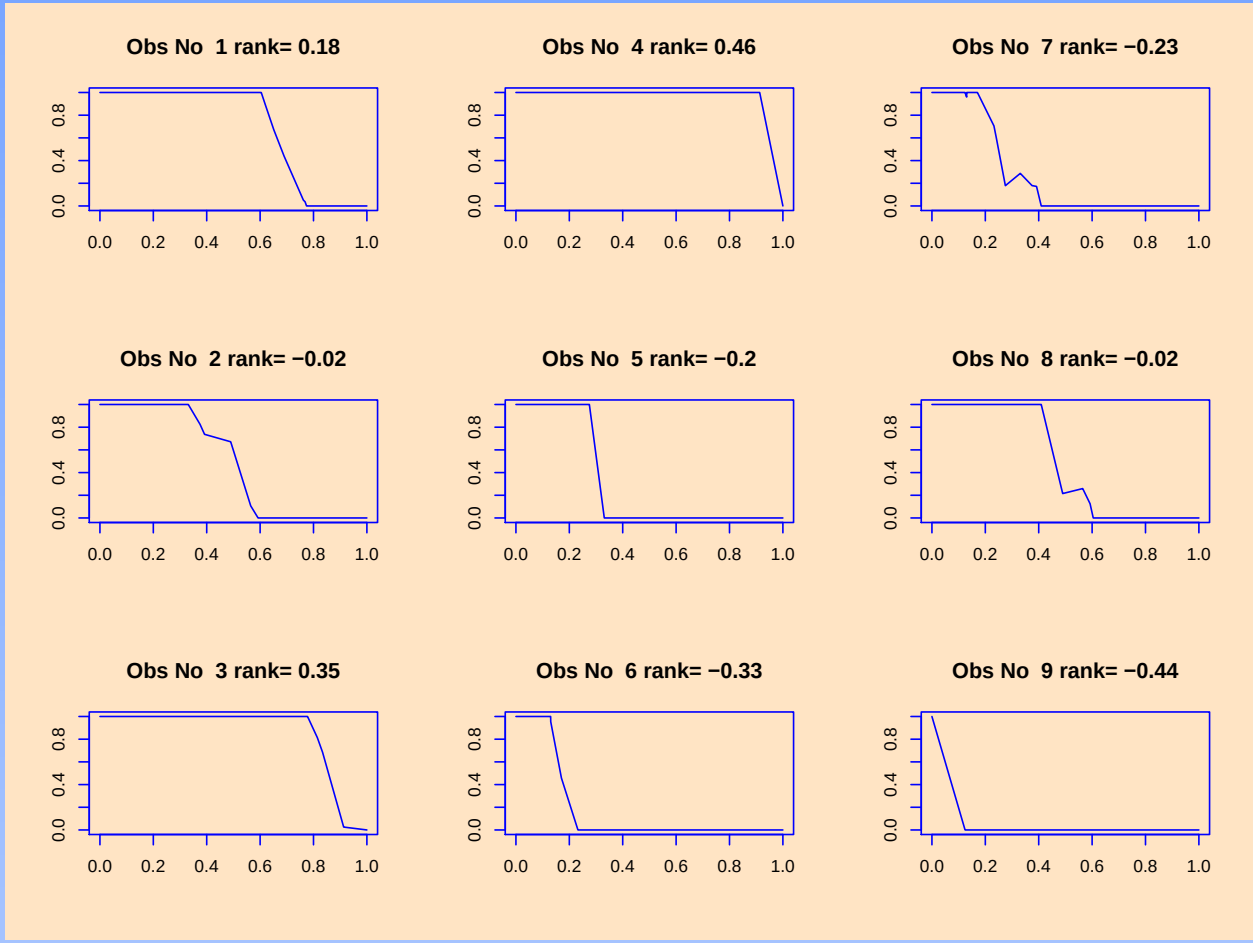
Theorem: (Gutenbrunner, Jurečková , Koenker and Portnoy) Under H_n and regularity conditions, the test statistic $T_n = S_n' Q_n^{-1} S_n$ where $S_n = (Z - \hat{Z})'\hat{b}_n, \quad \hat{Z} = X(X'X)^{-1}X'Z$, $Q_n = n^{-1}(Z - \hat{Z})'(Z - \hat{Z})$

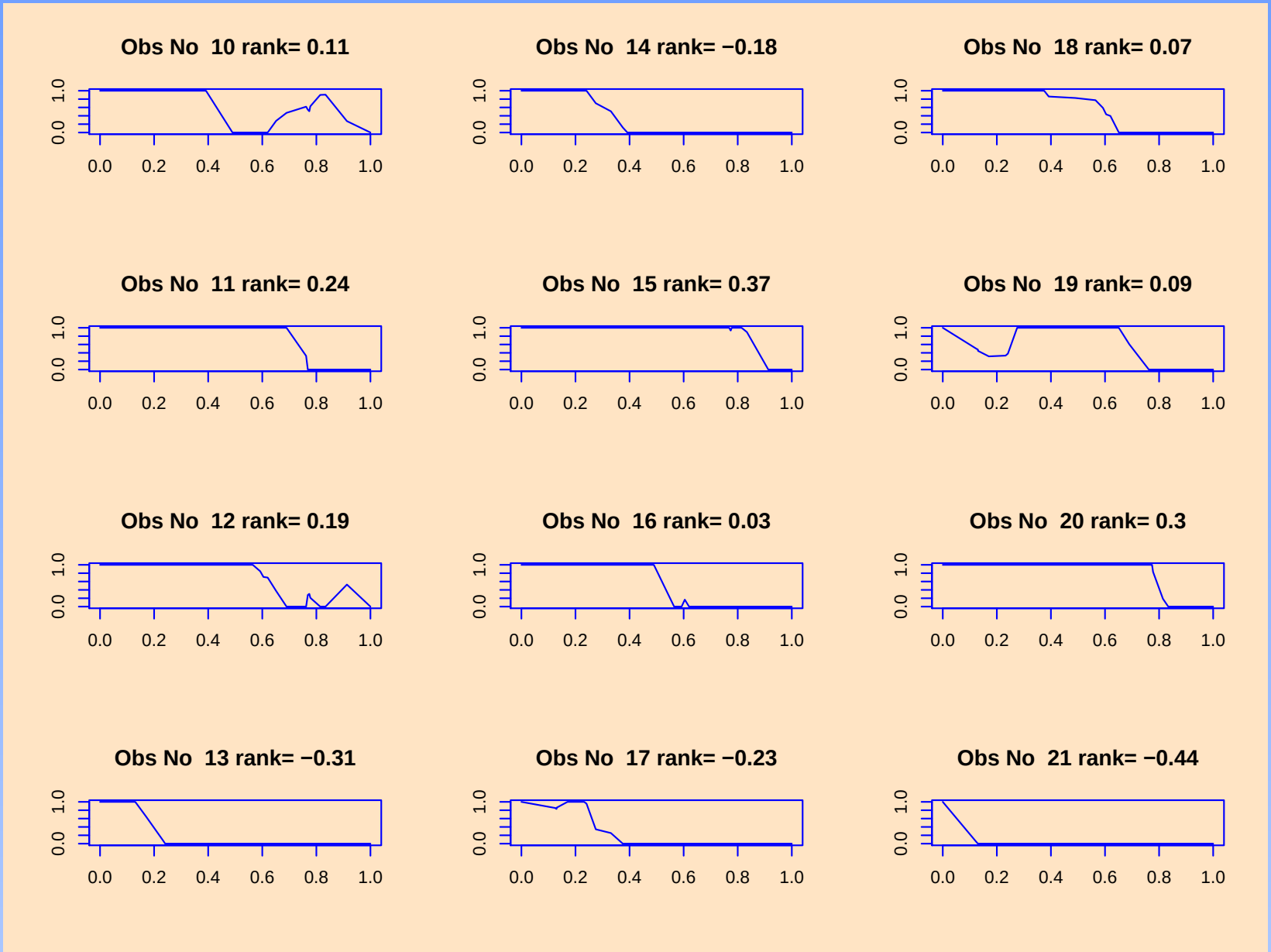
$$T_n \rightsquigarrow \chi_q^2(\eta)$$

where

$$\begin{aligned} \eta^2 &= \omega^2(\varphi, F)\gamma_0' Q \gamma_0 \\ \omega(\varphi, F) &= \int_0^1 f(F^{-1}(t))\, d\varphi(t) \end{aligned}$$

Regression Rankscores for Stackloss Data




Obs No 10 rank= 0.11
Obs No 14 rank= −0.18
Obs No 18 rank= 0.07
Obs No 11 rank= 0.24
Obs No 15 rank= 0.37
Obs No 19 rank= 0.09
Obs No 12 rank= 0.19
Obs No 16 rank= 0.03
Obs No 20 rank= 0.3
Obs No 13 rank= −0.31
Obs No 17 rank= −0.23
Obs No 21 rank= −0.44
0.0
1.0
0.0 0.2 0.4 0.6 0.8 1.0

Inversion of Rank Tests for Confidence Intervals

For the scalar γ case and using the score function

$$\varphi_\tau(t) = \tau - I(t < \tau)$$

$$\hat{b}_{ni} = -\int_0^1 \varphi_\tau(t) d\hat{a}_{ni}(t) = \hat{a}_{ni}(\tau) - (1-\tau)$$

where $\bar{\varphi} = \int_0^1 \varphi_\tau(t)dt = 0$ and $A^2(\varphi_\tau) = \int_0^1 (\varphi_\tau(t) - \bar{\varphi})^2 dt = \tau(1-\tau)$. Thus, a test of the hypothesis $H_0 : \gamma = \xi$ may be based on $\hat{a}_n$ from solving,

$$\max\{(y - x_2\xi)'a | X_1'a = (1-\tau)X_1'1, a \in [0,1]^n\} \tag{1}$$

and the fact that

$$S_n(\xi) = n^{-1/2} x_2' \hat{b}_n(\xi) \rightsquigarrow \mathcal{N}(0, A^2(\varphi_\tau) q_n^2) \tag{2}$$

Inversion of Rank Tests for Confidence Intervals

That is, we may compute

$$T_n(\xi) = S_n(\xi)/(A(\varphi_\tau)q_n)$$

where $q_n^2 = n^{-1}x_2'(I - X_1(X_1'X_1)^{-1}X_1')x_2$. and reject H_0 if $|T_n(\xi)| > \Phi^{-1}(1-\alpha/2)$.

Inverting this test, that is finding the interval of ξ's such that the test fails to reject. This is a quite straightforward parametric linear programming problem and provides a simple and effective way to do inference on individual quantile regression coefficients. Unlike the Wald type inference it delivers asymmetric intervals.

Inference on the Quantile Regression Process

Using the quantile score function, $\varphi_\tau(t) = \tau - I(t < \tau)$ we can consider the quantile rankscore process,

$$T_n(\tau) = S_n(\tau)' Q_n^{-1} S_n(\tau)/(\tau(1-\tau)).$$

where

$$\begin{aligned}
S_n &= n^{-1/2}(X_2 - \hat{X}_2)'\hat{b}_n, \\
\hat{X}_2 &= X_1(X_1'X_1)^{-1}X_1'X_2, \\
Q_n &= (X_2 - \hat{X}_2)'(X_2 - \hat{X}_2)/n, \\
\hat{b}_n &= (-\int \varphi(t) d\hat{a}_{in}(t))_{i=1}^n,
\end{aligned}$$

Inference on the Quantile Regression Process

Theorem: (Koenker and Machado) Under $H_n : \gamma(\tau) = O(1/\sqrt{n})$ for $\tau \in (0,1)$ the process $T_n(\tau)$ converges to a non-central Bessel process of order $q = \dim(\gamma)$

Related Wald and LR statistics can be viewed as providing a general apparatus for testing goodness of fit for quantile regression models. This approach is closely related to classical p-dimensional goodness of fit tests introduced by Kiefer (1959).

Quantile Regression Transformation Models

Given the transformation model the conditional quantile functions of $h(T)$, for $0 < \tau < 1$, are

$$Q_{h(T)}(\tau|x) = x^\top \beta + F_u^{-1}(\tau)$$

Since $P(h(T) \leq t) = P(T \leq h^{-1}(t))$, (monotone equivariance!)

$$Q_T(\tau|x) = h^{-1}(x^\top \beta + F_u^{-1}(\tau)).$$

Instead, we will consider,

$$Q_{h(T)}(\tau|x) = x^\top \beta(\tau),$$

for example, consider the location-scale shift model,

$$h(T_i) = x_i^\top \alpha + (x_i \gamma) u_i$$

with u_i iid from F. In this model we have a linear family of conditional quantile functions

$$Q_{h(T)}(\tau|x) = x^\top \alpha + (x^\top \gamma) F_u^{-1}(\tau) = x^\top \beta(\tau)$$

This is considerably more flexible.

Example: Cox Proportional Hazard Model

For the Cox model

$$\log \Lambda_0(T) = x^\top \beta + u$$

with $G(u) = 1 - \exp(-\exp(u))$. For Λ_0 Weibull,

$$\log \Lambda_0(t) = \gamma \log t - \alpha,$$

we obtain the accelerated failure time model,

$$\log T = x^\top \beta + u.$$

with iid u distributed as Weibull.

An Inference Problem

We would like to test whether covariates have a pure location shift effect on the response, a location-scale shift effect, or if they have some more general effect on the response distribution:

- Location Shift Hypothesis:

$$H_0 : \beta_i(\tau) = \alpha_i \quad i = 2, \ldots, p.$$

- Location-Scale Shift Hypothesis:

$$H_0 : \beta_i(\tau) = \alpha_i + \gamma_i \beta_1(\tau) \quad i = 2, \ldots, p.$$

Tests of the Kolmogorov-Smirnov type based on the whole quantile regression process will be considered.

The Kolmogorov-Smirnov Test

Suppose $\{Y_1, \ldots, Y_n\}$ are *iid* from df F. We would like to test,

$$H_0 : F = F_0.$$

We want to consider the K-S statistic,

$$K_n = \sup_{x \in \mathbb{R}} \sqrt{n} |F_n(x) - F_0(x)|$$

where $F_n(x) = n^{-1} \sum I(Y_i \leq x)$.

KS Test is ADF

Classically, from Doob (1949), we know

$$U_n(x) = \sqrt{n}(F_n(x) - F_0(x))$$

or, changing variables $x \longrightarrow F_n^{-1}(\tau)$,

$$u_n(\tau) = \sqrt{n}(\tau - F_0(F_n^{-1}(\tau)))$$

converges weakly under H_0 to a Brownian Bridge process, i.e., a Gaussian process, u_0, with mean zero and covariance function $Cov(u_0(\tau_1), u_0(\tau_2)) = \tau_1 \wedge \tau_2 - \tau_1\tau_2$. so the test is *asymptotically distribution free* (ADF).

The Durbin Problem

Now suppose F_0 is known only up to parameters, e.g., $F_0(x, \theta_0) = \Phi((x - \mu_0)/\sigma_0)$, but $\theta_0 = (\mu_0, \sigma_0)$ is unknown. We are tempted to consider the process,

$$\hat{U}_n(x) = \sqrt{n}(F_n(x) - F_0(x, \hat{\theta}_n))$$

and again changing variables, so $\tau = F_0(x, \theta_0)$, setting $G(\tau, \theta_0) = \tau$,

$$\hat{u}_n(\tau) = \sqrt{n}(G_n(\tau) - G(\tau, \hat{\theta}_n))$$

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$$\hat{u}_n(\tau) = \sqrt{n}(G_n(\tau) - G(\tau, \hat{\theta}_n))$$

Like $u_n(\tau)$, $\hat{u}_n(\tau)$ converges weakly to zero mean Gaussian process, say, $\hat{u}_n(\tau) \Rightarrow \hat{u}_0(\tau)$, but now for the mle $\hat{\theta}_n$,

$$E(\hat{u}_0(\tau_1)\hat{u}_0(\tau_2)) = \tau_1 \wedge \tau_2 - \tau_1\tau_2 - g_0(\tau_1)^\top \mathcal{J}^{-1} g_0(\tau_2)$$

where $g_0(\tau) = \partial F_0(y, \theta_0)/\partial\theta|_{y=F_0^{-1}(\tau,\theta_0)}$, and $\mathcal{J}$ is the Fisher information about θ in model F_0. Now $\hat{K}_n = \sup|\hat{u}_n(\tau)|$ depends on F_0; this is *the Durbin Problem*.

The Doob-Meyer Decomposition

The process $G_n(\tau) = F_0(F_n^{-1}(\tau))$ is Markov:

$$n\Delta G_n(\tau) = n[G_n(\tau + \Delta\tau) - G_n(\tau)] \sim \text{Bin}(n(1 - G_n(\tau)), \Delta\tau/(1 - \tau)).$$

$$E[\Delta G_n(\tau)|\mathcal{F}_\tau^{G_n}] = \frac{1 - G_n(\tau)}{1 - \tau}\Delta\tau$$

and this *suggests* the representation,

$$G_n(t) = \int_0^t \frac{1 - G_n(s)}{1 - s} ds + m_n(t)$$

where $m_n(t)$ is a martingale. Now substituting from $u_n(t) = \sqrt{n}(G_n(t) - t)$ we have

$$w_n(t) = u_n(t) + \int_0^t \frac{u_n(s)}{1 - s} ds$$

where $w_n(t) = \sqrt{n} m_n(t) \Rightarrow w_0(\tau)$, is standard Brownian motion.

"Marmalade" in a Martingale

Etymology: [OED] a. Fr. martingale of obscure etymology. [First found in Rabelais in *chausses a la martingale*, men's socks that fastened at the back of the leg. This is commonly supposed to mean literally 'hose after the fashion of Martigues' (in Provence).

Doob-Meyer as Recursive OLS

Let $g(t) = (t, g_1(t), \ldots, g_p(t))^\top$ be a $(p+1)$-vector of real-valued functions on $[0,1]$. Suppose $\dot{g}(t) = dg(t)/dt$ are linearly independent, so

$$C(t) = \int_t^1 \dot{g}(s)\dot{g}(s)^\top ds$$

is nonsingular, and consider the transformation,

$$w_n(t) = v_n(t) - \int_0^t \dot{g}(s)^\top C^{-1}(s) \int_s^1 \dot{g}(r) dv_n(r) ds$$

In the Doob-Meyer case, we set $g(t) = t$ so $\dot{g}(t) = 1$, $C(s) = 1 - s$, and noting that,

$$\int_s^1 \dot{g}(r) dv_n(r) = v_n(1) - v_n(s) = -v_n(s)$$

we obtain the Doob-Meyer decomposition.

Khmaladze's Martingalization

Ingredients:

$$G(\tau, \hat{\theta}_n) = \tau + (\hat{\theta} - \theta_0)^\top g(\tau, \theta^*)$$
$$\sqrt{n}(\hat{\theta} - \theta_0) = \int_0^1 h(s, \theta_0) du_n(s) + o_p(1)$$
$$\hat{u}_n(\tau) = \sqrt{n}(G_n(\tau) - \tau + \tau - G(\tau, \hat{\theta}_n))$$

Combine and stir:

$$\hat{u}_n(\tau) = u_n(\tau) - g(\tau, \theta_0)^\top \int_0^1 h(s, \theta_0) du_n(s) + o_p(1) \quad (3)$$
$$\Rightarrow u_0(\tau) - g(\tau, \theta_0)^\top \int_0^1 h(s, \theta_0) du_0(s) \quad (4)$$

$$\tilde{u}_n(\tau) = \hat{u}_n(\tau) - \int_0^\tau \dot{g}(s)^\top C^{-1}(s) \int_s^1 \dot{g}(r) d\hat{u}_n(r) ds \quad (5)$$
$$\Rightarrow w_0(\tau) \quad (6)$$

Martingalization annihilates the $g(\tau, \theta_0)$ term and restores ADF property of KS-test!

Khmaladze for the Quantile Process

Let

$$\hat{\alpha}(\tau) = \text{argmin}_{a \in \mathbb{R}} \sum \rho_\tau(y_i - a)$$

where $\{y_i\}$ are iid from $F_0((y-\mu)/\sigma)$. Consider

$$H_0: \quad \alpha(\tau) = F_y^{-1}(\tau) = \mu + \sigma F_0^{-1}(\tau)$$

under H_0,

$$v_n(\tau) = \sqrt{n}\varphi_0(\tau)(\hat{\alpha}(\tau) - \alpha(\tau))/\sigma \Rightarrow v_0(\tau)$$

where $\varphi_0(\tau) = f_0(F_0^{-1}(\tau))$ and $v_0(\tau)$ is the Brownian Bridge process.
To test H_0, set $\tilde{\alpha}(\tau) = \xi(\tau)^\top \tilde{\theta} = (1, F_0^{-1}(\tau))\tilde{\theta}$, then

$$\hat{v}_n(t) = \sqrt{n}\varphi_0(t)(\hat{\alpha}(t) - \tilde{\alpha}(t))/\sigma \tag{7}$$

$$= \sqrt{n}\varphi_0(t)(\hat{\alpha}(t) - \alpha(t) - (\tilde{\alpha}(t) - \alpha(t)))/\sigma \tag{8}$$

$$= v_n(t) - \sqrt{n}\varphi_0(\tau)(\tilde{\theta} - \theta_0)^\top \xi(t)/\sigma \tag{9}$$

Testing for Normality

In the typical case that θ_0 consists of a location and scale parameter we have,

$$g(\tau) = (\tau, \varphi_0(\tau)\xi(\tau)^\top)^\top$$

so,

$$\dot{g}(\tau) = (1, \dot{f}/f, 1 - F_0^{-1}(\tau)\dot{f}/f)^\top$$

where $\dot{f}/f$ is evaluated at $F^{-1}(\tau)$. In the Gaussian case, $F_0 = \Phi$, we have

$$\dot{g}(\tau) = (1, -\Phi^{-1}(\tau), 1 - \Phi^{-1}(\tau)^2)^\top$$

Inference for Quantile Regression

Now consider the quantile regression process,

$$\hat{\beta}(\tau) = \text{argmin}_{b \in \mathbb{R}^p} \sum \rho_\tau(y_i - x_i^\top b)$$

The analogue of the location scale model is

$$y_i = x_i^\top \alpha + (x_i^\top \gamma) u_i$$

with $\{u_i\}$ iid from F_0. This implies the null hypothesis,

$$H_0 : \quad \beta_i(\tau) = \alpha_i + \gamma_i F_0^{-1}(\tau) \quad i = 1, ..., p.$$

We would like to test, H_0, versus a general alternative. Note that, H_0 implies that all p coordinates of $\beta(\cdot)$ are affine functions of a single univariate function.

Simple Nulls

When α, γ, F_0 are all known we have, subject to some regularity conditions,

$$v_n(\tau) = \sqrt{n} J_n^{-1/2} H_n(\hat{\beta}(\tau) - \beta(\tau)) \Rightarrow v_0$$

where v_0 is now a p-variate Brownian Bridge, $J_n = n^{-1} X^\top X$, $H_n = n^{-1} X^\top \Gamma^{-1} X$, and $\Gamma = \text{diag}(x_i^\top \gamma)$.

This leads to Wald, LR and LM/rankscore tests as in Koenker and Machado (JASA, 1999), employing Bessel processes as in Kiefer(1959). But when (α, γ) are unknown, the Durbin problem arises again.

A General Linear Hypothesis

Consider the hypothesis,

$$R\beta(\tau) - r = \Psi(\tau) \qquad \tau \in \mathcal{T} \tag{10}$$

where R denotes a $q \times p$ matrix, $q \leq p, r \in \mathbb{R}^q$, and $\Psi(\tau)$ denotes a known function $\Psi : \mathcal{T} \to \mathbb{R}^q$. and the local alternative,

$$R\beta_n(\tau) - r - \Psi(\tau) = \zeta(\tau)/\sqrt{n}.$$

Test based on:

$$v_n(\tau) = \sqrt{n}\varphi_0(\tau)(R\Omega R^\top)^{-1/2}(R\hat{\beta}(\tau) - r - \Psi(\tau))$$

where $\Omega = H_0^{-1} J_0 H_0^{-1}$ with $J_0 = \lim n^{-1} \sum x_i x_i^\top$, and $H_0 = \lim n^{-1} \sum x_i x_i^\top / \gamma^\top x_i$.

Regularity Conditions

Assumption 1. *The distribution function F_0, has a continuous Lebesgue density, f_0, with $f_0(u) > 0$ on $\{u : 0 < F_0(u) < 1\}$.*

Assumption 2. *The sequence of design matrices $\{X_n\} = \{(x_i)_{i=1}^n\}$ satisfy:*

(i) $x_{i1} \equiv 1 \qquad i = 1, 2, \ldots$
(ii) $J_n = n^{-1} X_n^\top X_n \to J_0$, *a positive definite matrix.*
(iii) $H_n = n^{-1} X_n^\top \Gamma_n^{-1} X_n \to H_0$, *a positive definite matrix where* $\Gamma_n = diag(\gamma^\top x_i)$.

Assumption 3. *There exists a fixed, continuous function $\zeta(\tau) : [0, 1] \to \mathbb{R}^q$ such that for samples of size n,*

$$R\beta_n(\tau) - r - \Psi(\tau) = \zeta(\tau)/\sqrt{n}.$$

More Regularity Conditions

Assumption 4. *There exist estimators $\varphi_n(\tau)$ and Ω_n satisfying*

i. $\sup_{\tau \in \mathcal{T}} |\varphi_n(\tau) - \varphi_0(\tau)| = o_p(1)$,
ii. $||\Omega_n - \Omega|| = o_p(1)$.

Assumption 5. *The function $g(t)$ satisfies:*

i $\int \parallel \dot{g}(t) \parallel^2 dt < \infty$,
ii $\{\dot{g}_i(t) : i = 1, \ldots, m\}$ *are linearly independent in a neighborhood of 1.*

Theorem 1. *Let $\mathcal{T}$ denote the closed interval $[\varepsilon, 1-\varepsilon]$, for some $\varepsilon \in (0, 1/2)$. Under conditions A.1-3*

$$v_n(\tau) = \sqrt{n}\varphi_0(\tau)(R\Omega R^\top)^{-1/2}(R\hat{\beta}(\tau) - r - \Psi(\tau)) \tag{11}$$

$$\Rightarrow v_0(\tau) + \eta(\tau) \text{ for } \tau \in \mathcal{T} \tag{12}$$

where $v_0(\tau)$ denotes a q-variate standard Brownian bridge process and $\eta(\tau) = \varphi_0(\tau)(R\Omega R^\top)^{-1/2}\zeta(\tau)$. Under the null hypothesis, $\zeta(\tau) = 0$, the test statistic

$$\sup_{\tau\in\mathcal{T}} \| v_n(\tau) \| \Rightarrow \sup_{\tau\in\mathcal{T}} \| v_0(\tau) \| .$$

Theorem 2. *Under conditions A.1-5, we have*

$$\hat{v}_n(\tau) = \sqrt{n}\varphi_0(\tau)[R_n\Omega R_n^\top]^{-1/2}(R_n\hat{\beta}(\tau) - r_n - \Psi(\tau)) \tag{13}$$

$$\Rightarrow Z_n^\top \xi(\tau) + v_0(\tau) + \eta(\tau) \text{ for } \tau \in \mathcal{T} \tag{14}$$

where $\xi(\tau) = \varphi_0(\tau)(1, F_0^{-1}(\tau))^\top$, and $Z_n = \mathcal{O}_p(1)$, with $v_0(\tau)$ and $\eta(\tau)$ as specified in Theorem 1.

Theorem 3. *Under conditions A.1 - 6, we have*

$$\tilde{v}_n(\tau)^\top = \hat{v}_n(\tau)^\top - \int_0^\tau \dot{g}(s)^\top C^{-1}(s) \int_s^1 \dot{g}(r) d\hat{v}_n(r)^\top ds \tag{15}$$

$$\Rightarrow w_0(\tau) + \tilde{\eta}(\tau) \text{ for } \tau \in \mathcal{T} \tag{16}$$

where $w_0(\tau)$ *denotes a* q*-variate standard Brownian motion, and under the null hypothesis,* $\zeta(\tau) = 0$,

$$\sup_{\tau \in \mathcal{T}} \| \tilde{v}_n(\tau) \| \Rightarrow \sup_{\tau \in \mathcal{T}} \| w_0(\tau) \| .$$

Pennsylvania Reemployment Bonus Experiment

Can the durations of insured unemployment spells be shortened by offering cash bonuses to recipients for early reemployment?

- 1988-89 Experiment in Pennsylvania
- 6 Treatments + Control Group
  - ⋆ Two levels of bonus payment
  - ⋆ Two settings of the qualification period
- Randomized Assignment to Groups
- 13,913 Participants

Experimental Treatment Groups

Table 1: Treatment Groups

Group	Bonus Amount	Qualification Period	Workshop Offer
Controls	0	0	No
Treatment 1	Low	Short	Yes
Treatment 2	Low	Long	Yes
Treatment 3	High	Short	Yes
Treatment 4	High	Long	Yes
Treatment 5	Declining	Long	Yes
Treatment 6	High	Long	No

Note: The low benefit was 3 times UI weekly benefit amount, the high benefit was 6 times this amount. The declining bonus declined from 6 times the weekly benefit to zero, over a 12 week period. The short qualification period was 6 weeks, and the long period was 12 weeks.

Sample Sizes

Groups	Target n	Collected n	Analysis n
Control	3,000	3,392	3,354
Treatment 1	1,030	1,395	1,385
Treatment 2	2,240	2,456	2,428
Treatment 3	1,740	1,910	1,885
Treatment 4	1,590	1,771	1,745
Treatment 5	1,740	1,860	1,831
Treatment 6	1,780	1,302	1,285
Total	13,120	14,086	13,913

Duration Histograms

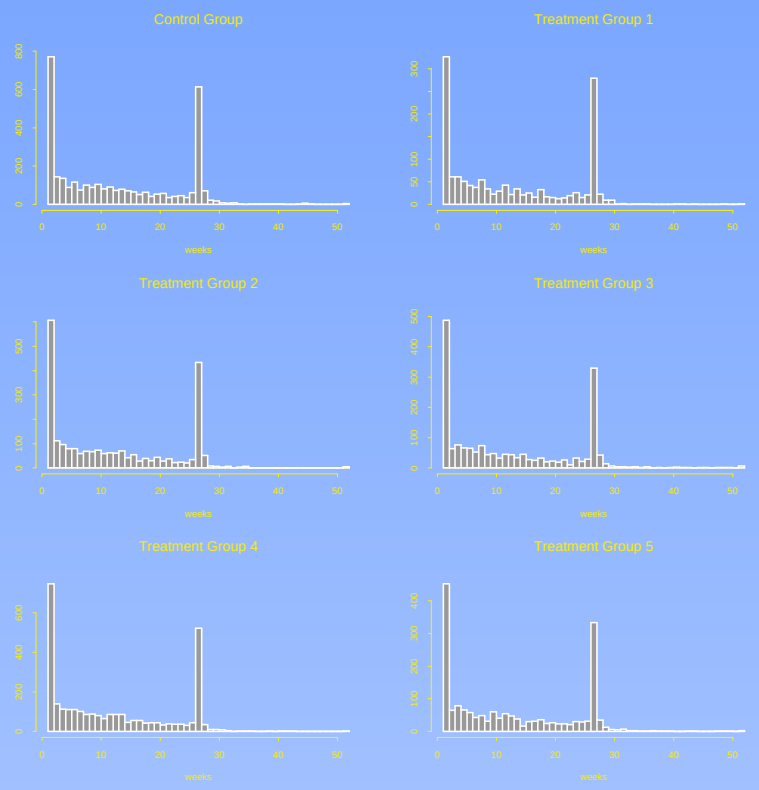
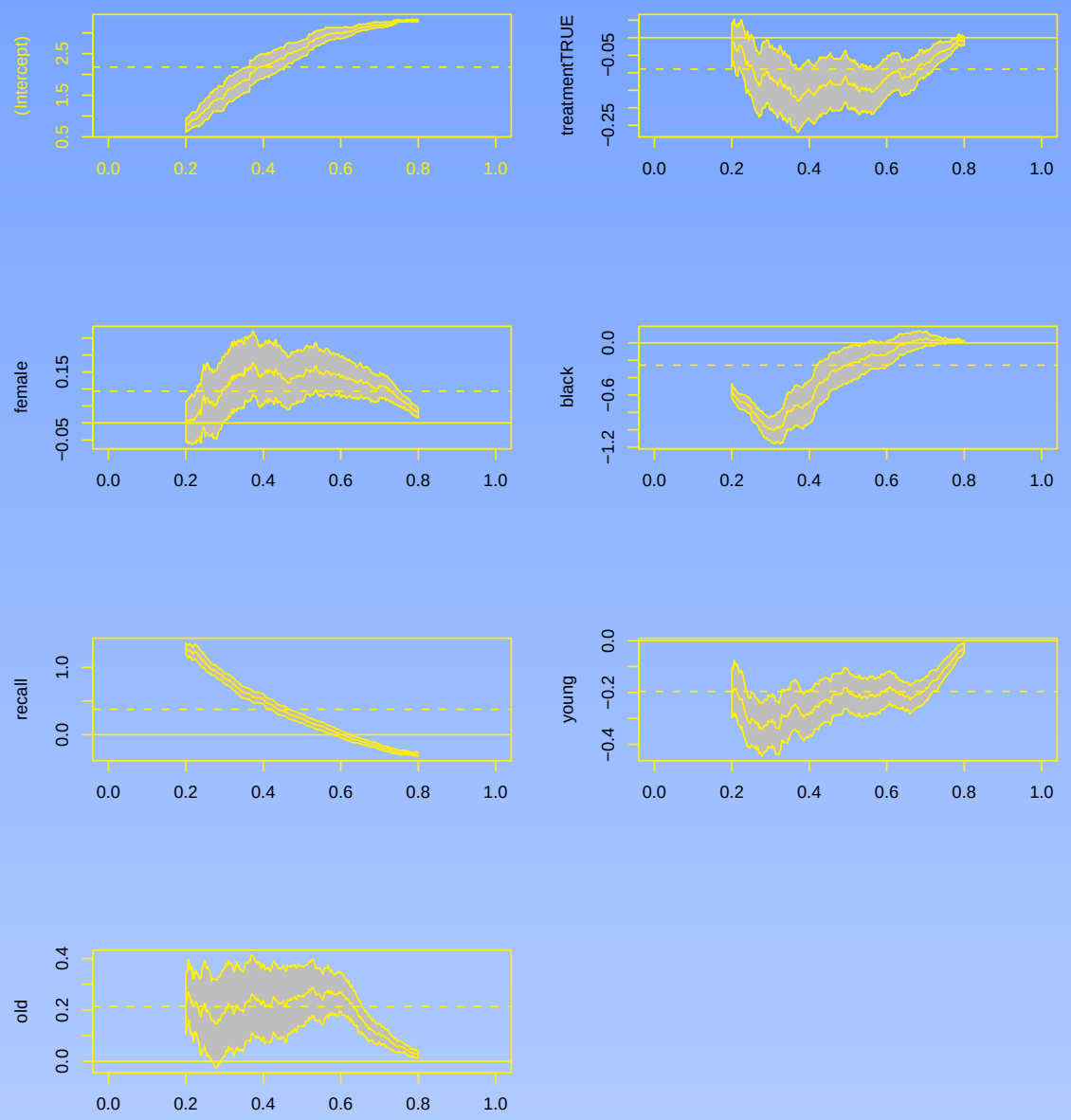
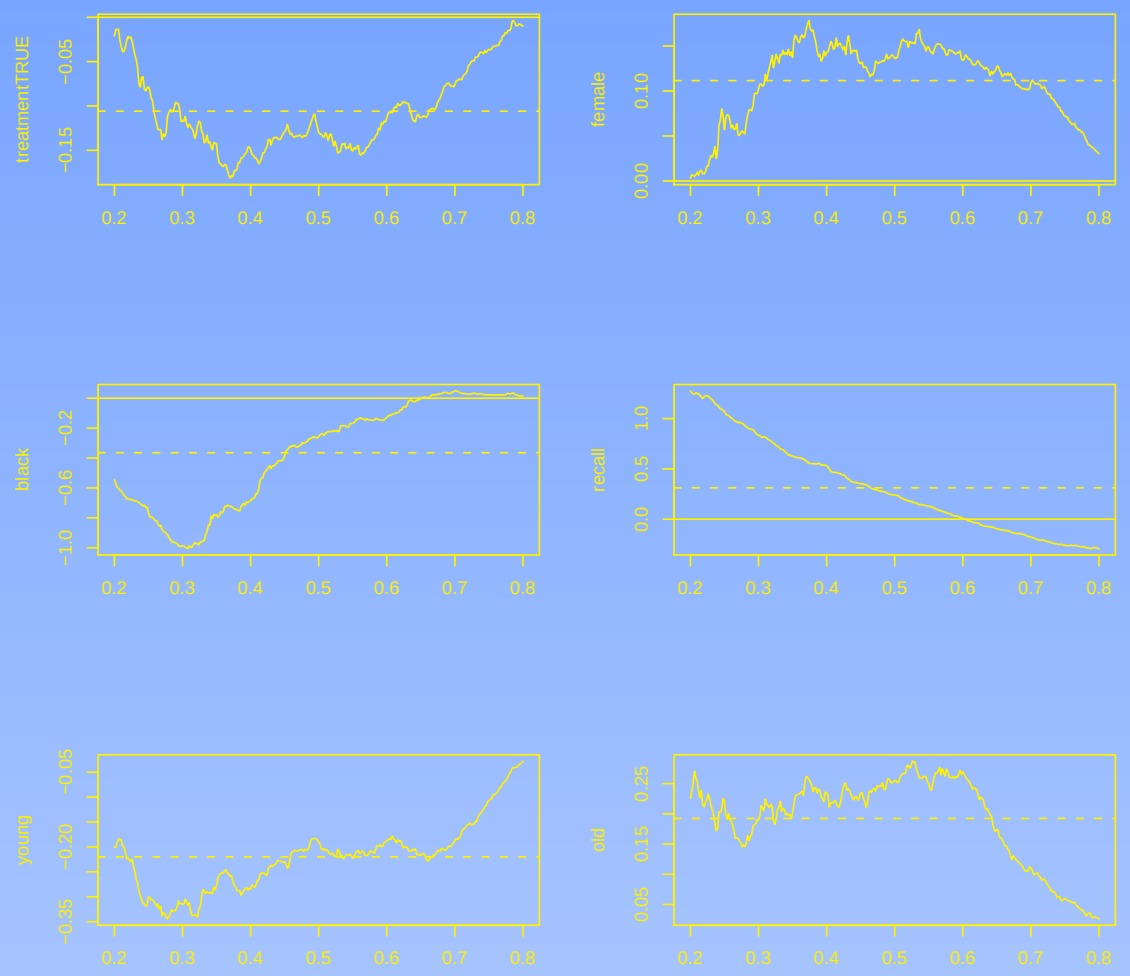


Figure 1: Duration (in weeks) of UI benefits by treatment.

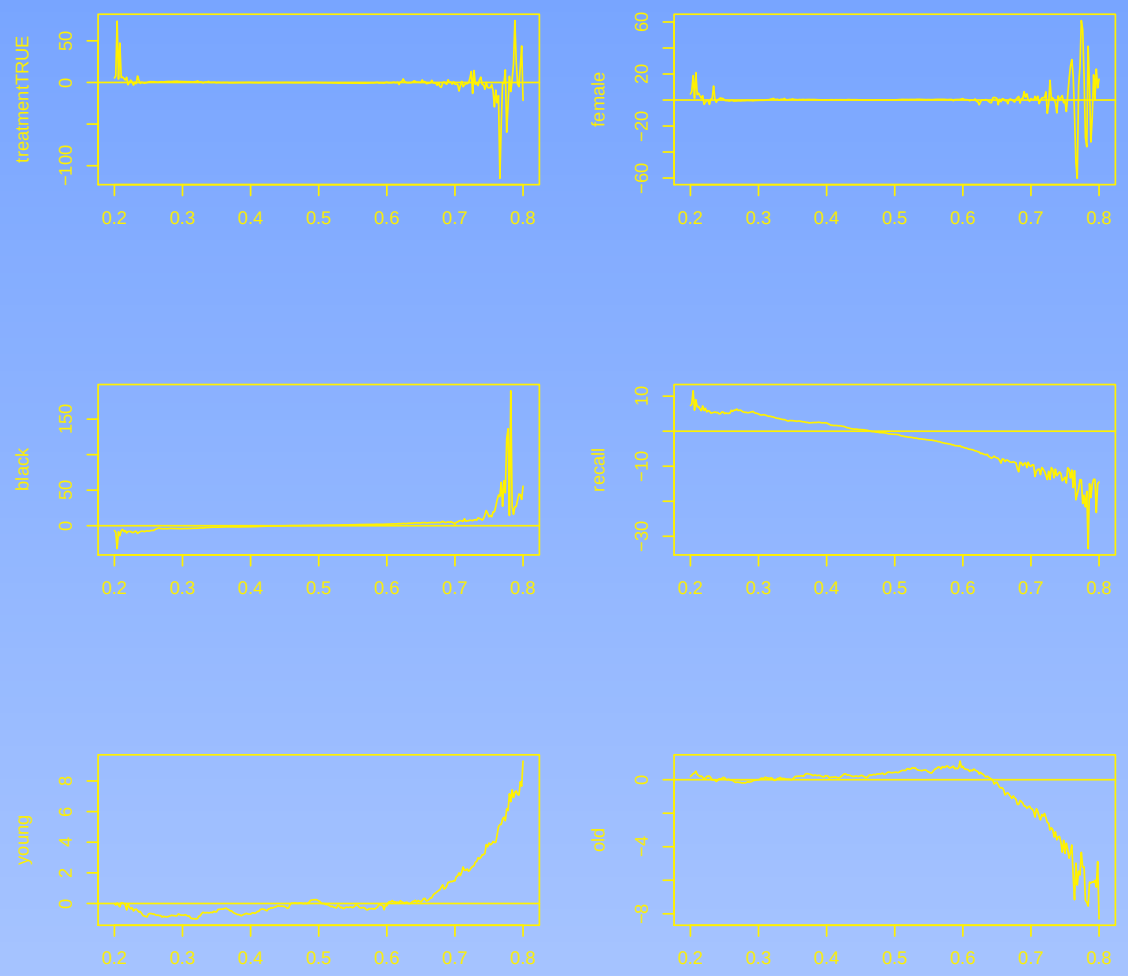
Quantile Regression Process



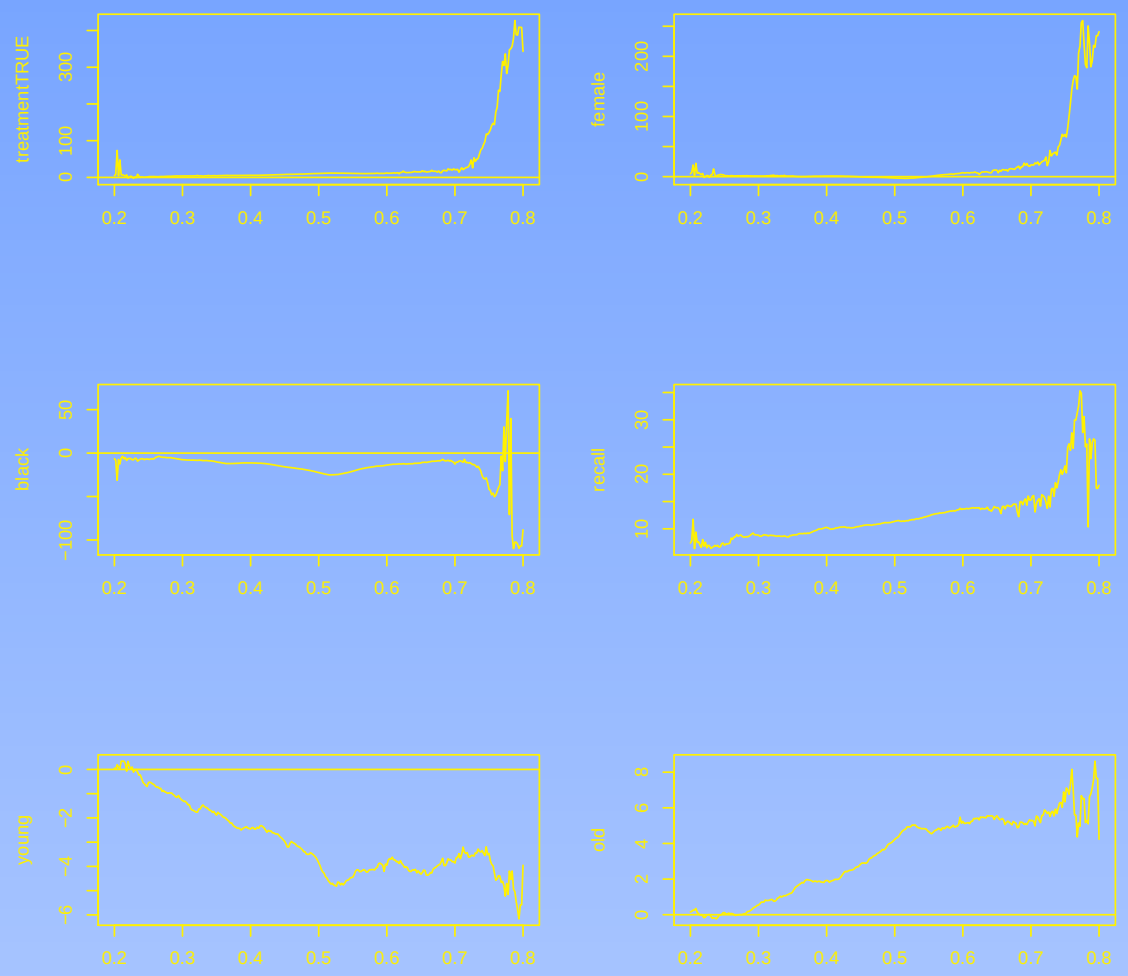
Fitted Quantile Regression Process



Standardized Residual Quantile Regression Process



Khmaladzized Quantile Regression Process



Test Results

Variable	Location Scale Shift	Location Shift
Treatment	5.41	5.48
Female	4.47	4.42
Black	5.77	22.00
Hispanic	2.74	2.00
N-Dependents	2.47	2.83
Recall Effect	4.45	16.84
Young Effect	3.42	3.90
Old Effect	6.81	7.52
Durable Effect	3.07	2.83
Lusd Effect	3.09	3.05
Joint Effect	112.23	449.83

Table 2: Tests of the Location-Scale and Location Shift Hypotheses: Critical values for the univariate tests are 1.92 at .05 and 2.42 at .01. For the joint tests the .01 critical value is 16.0.

Some Concluding Comments about Inference

- Asymptotic inference for quantile regression poses some statistical challenges since it involves elements of nonparametric density estimation.
- Classical rank statistics and Hájek 's rankscore process are closely linked via Gutenbrunner and Jurečková 's regression rankscore process.
- Inference on the quantile regression process can be conducted with the aid of Khmaladze's extension of the Doob-Meyer construction.
- Resampling offers many further lines of development for inference in the quantile regression setting.